Friedrich-Alexander-Universität
Technische Fakultät

Pattern Recognition Lab

FAU

Introduction to Software Engineering

Exercise 2: Software Processes

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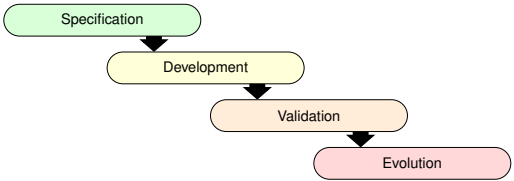
1. Recap
2. First task
3. Second task
4. Third task
5. Fourth task

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Recap

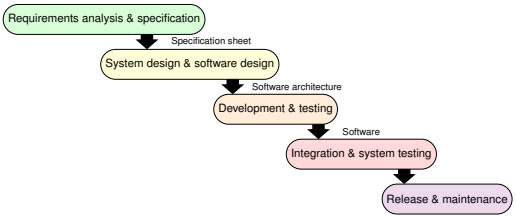
Software Processes and Models

Software processes



Source: Sommerville 2016 [1] p. 44 (adapted)

Waterfall Model

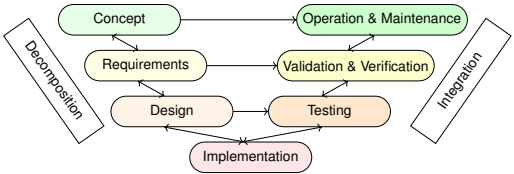


Source: Sommerville 2016 [1] p. 47 (adapted)

Recap

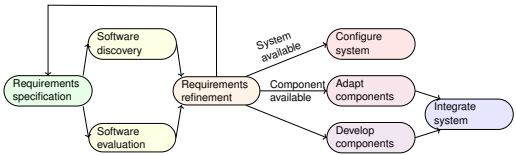
Software Processe Models

V-Model



Source: Stephens 2015 [2] p. 276 (adapted)

Reuse-oriented approach

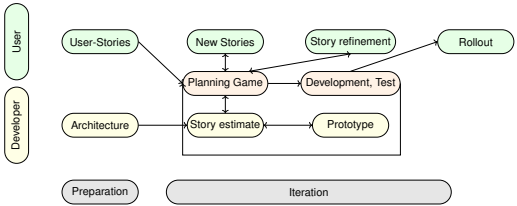


Source: Sommerville 2016 [1] p. 52 (adapted)

Recap

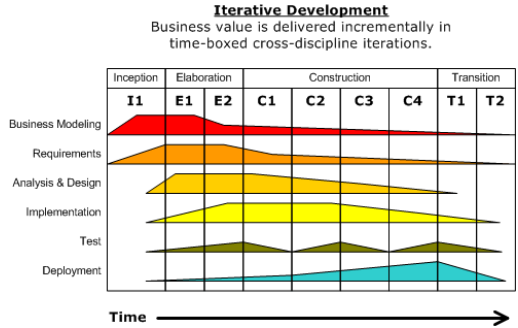
Agile Models

Agile model



Source: Metzner 2020 [3] p. 26 (adapted)

Rational Unified Process (RUP)



Source: https://commons.wikimedia.org/w/index.php?curid=37249677

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Software Processes

First task

Your boss gives your team a new project:

> A Python class is required that performs basic mathematical operations (addition, subtraction, multiplication, division) with two variables. This class is to be integrated into an existing project on the company server that handles string operations. All string operations are in the file string_manipulation.py, while all mathematical operations are to be in the file math.py. Due to the manageability of this class, only a handful of functional tests are necessary after implementation.

Questions:

1. Is the Waterfall model well suited for this project?
2. To present the project plan to your boss, fill out the provided Rational Unified Process (RUP) diagram for the given project when managed using the waterfall model.

1. Recap
2. First task
3. **Second task**
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Software Processes

Second task

After you've worked on the project for a while, developing part of the code, your boss calls for another meeting. He gives you new additional requirements. The class should also multiply arrays with each other. Specifically, the code should work on these example inputs and give these outputs:

Input	Add	Sub	Mul	Div
3,4	7	-1	12	0.75
5, 0	5	5	0	Inf
-1, 2	1	-3	-2	-0.5
'a', 7	Exception	Exception	Exception	Exception
(2 +6i), (4-2i)	6+4i	-2 + 8i	20+20i	-0.2 + 1.4i
[6,3,8] , [12,3,10]	[18, 6, 18]	[-6, 0, -2]	[72, 9, 80]	[0.5, 1, 0.8]

Questions:

1. What issues could have arisen while planning and implementing this project with the Waterfall model, especially due to these new requirements?
2. Fill in the RUP Diagram again, this time for the changed project plan. Compare how the RUP Diagram would look like for the iterative and incremental Waterfall Models.

Software Processes

Second task (cont.)

Questions:

3. Illustrate the V-Model within a RUP diagram, specifically for this project where the requirements have been changed and briefly explain your solution.
4. How would the project planning be different with the V-model? Would it mitigate these issues?

	Inception	Elaboration	Construction	Transition
Business Modeling				
Requirements				
Analysis & Design				
Implementation				
Testing				
Deployment				

1. Recap
2. First task
3. Second task
4. **Third task**
5. Fourth task

Software Processes

Third task

Shortly after the meeting, your boss walks up to your desk.

He is experiencing pressure from management and asks you to spend as little time as possible on this task in order to focus on more complex projects.

Questions:

1. Which approach could you choose to meet his expectations?
2. Explain how the chosen approach would be used and why you are choosing it. Are there any compromises?

1. Recap
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Software Processes

Fourth task

In order to manage our game code better, we want to use Git.

Follow the instructions to create a git repository for your game code.

1. Create a GitHub/GitLab account and a new project for the game
2. Push the current status of the game

Software Processes

Git

GitHub is a web platform to manage and collaborate on projects.

- Repository: A place where your project lives. It contains all project files, history, and issues.
- Commit: A snapshot of your project at a point in time.
- Branch: A parallel version of the code, often used for new features or bug fixes.
- Pull Request: A proposed change to the code that needs review and approval.

Git (cont.)

GitHub Workflow

- Create a Repository: Start a new project or use an existing one.
- Clone: Download a copy of the repository to your local machine.
- Make Changes: Edit code, add files, or make improvements.
- Commit: Save your changes and create a new commit.
- Push: Upload your commits to the remote repository on GitHub.
- Create a Pull Request: Propose your changes to be merged into the main codebase.
- Review and Collaborate: Team members review your changes, suggest improvements, and discuss.
- Merge: Once approved, your changes are merged into the main codebase.

Software Processes

Git (cont.)

Cloning

```
git clone https://github.com/username/my-project.git
```

Add changes

```
git add myfile.py
```

Commit

```
git commit -m "Added new feature"
```

Push

```
git push origin main
```

References I

[1] Ian Sommerville. *Software Engineering*. Always learning. Pearson, 2016.

[2] Rod Stephens. *Beginning Software Engineering*. Wrox beginning guides. Wiley, 2015.

[3] Anja Metzner. *Software Engineering - kompakt*. Carl Hanser Verlag GmbH & Company KG, 2020.